

SE LAB 1

Requirements Engineering & UML Use-Case Modelling

**Problem Statement #14 — Healthcare & Telemedicine
Corporate Mental Wellness & Counseling Platform**

Individual Use-Case Report: UC-03 — Book Counseling Session

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1. Scenario and Approach

The system is an anonymous counselling portal for a company. Employees can check their stress levels, book confidential sessions with a licensed therapist and follow their own wellness indicators. The employer does not learn who used the counselling service.

For this individual submission, the master report has been reduced to one core use case: UC-03 — Book Counseling Session. The other three detailed use-case flows have been removed. The requirements table remains complete because the lab handout requires exactly five functional requirements and two non-functional requirements.

2. Stated Assumptions

- The employee has a corporate identity used once to prove eligibility. The counselling side uses a randomly generated pseudo-ID such as WEL-7F42A91C.
- The pseudo-ID to employee mapping is stored separately from counselling records. Booking, assessment and notes data use the pseudo-ID rather than the corporate identity.
- Session delivery, therapist licensing checks and payment/billing are outside this Lab 1 iteration.

3. Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall verify an employee against the corporate directory once at registration, issue a randomly generated pseudo-ID, and use only that pseudo-ID in every record and therapist-facing screen.	High	Pass: booking record and therapist dashboard show only WEL-XXXXXXX; mapping is not readable from counselling data. Fail: email, employee ID or name appears, or pseudo-ID is derived from corporate identity.	Protects confidentiality and makes employees willing to use the service.
FR-002	Functional	The system shall let an employee complete a ten-item stress self-assessment using a five-point frequency scale and return the total score out of 40 with its band.	High	Pass: complete submission returns a 0–40 score and matching band within 2 seconds. Fail: missing result, contradictory band, or partial form is scored.	Provides a consistent wellness check.
FR-003	Functional	The system shall allow an employee to book a counselling session only in a slot the therapist has published as free,	High	Pass: concurrent attempts for one slot result in exactly one confirmed booking. Fail: double booking or	Prevents double booking.

		and shall mark that slot unavailable in the same transaction.		acceptance outside published availability.	
FR-004	Functional	The system shall show an employee a trend of their own wellness indicators across their six most recent assessments.	Medium	Pass: existing assessments are plotted in date order. Fail: another account's data appears or missing dates are filled with estimated values.	Makes changes in wellness easier to understand.
FR-005	Functional	Whenever an assessment falls in the severe band, the system shall display regional crisis helpline details and offer the earliest available priority appointment slot.	High	Pass: severe result shows helpline details and at least one priority slot within 24 hours. Fail: severe result is treated as an ordinary result.	Provides immediate support when required.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional (Performance & Security)	The appointment scheduler shall convert every slot to the viewer's local timezone correctly, including daylight-saving transitions, and return a slot list in under 2 seconds at peak load.	High	Pass: 500 concurrent users across IST, GMT and PST keep 95th-percentile slot-list latency below 2 seconds and show the same absolute appointment time. Fail: timezone disagreement or latency above 2 seconds.	Employees and therapists may be in different countries; timezone errors can cause missed sessions.
NFR-002	Nonfunctional (Security & Privacy)	The pseudo-ID mapping shall be stored separately from counselling data; mapping, notes and assessment answers shall use AES-256 at rest and TLS 1.3 in transit; aggregate reports require groups of at least ten employees.	High	Pass: counselling DB dump contains no readable notes or corporate identifiers, and groups of fewer than ten are rejected. Fail: plaintext data, readable mapping, or small-group report is released.	Protects sensitive counselling records and prevents re-identification.

4. Actors

Actor	Kind	Role in the System
Corporate Employee	Primary	Books and attends counselling sessions. The counselling side knows the employee

		only by pseudo-ID.
Licensed Therapist	Primary / Secondary for booking	Publishes availability and conducts sessions. The therapist sees the pseudo-ID, not the corporate identity.
Notification Service	Supporting	External email/push provider that delivers booking confirmations and reminders.

5. Selected Use Case and Traceability

Use Case	Title	Actor / Relationship	Traces To
UC-03	Book Counseling Session	Corporate Employee; Licensed Therapist (indirect); Notification Service	FR-003; NFR-001
UC-05	Manage Availability Calendar	Licensed Therapist	FR-003
UC-08	Generate Pseudo-ID	Included by UC-03	FR-001
UC-10	Verify Slot Availability	Included by UC-03	FR-003
UC-11	Send Confirmation Notification	Included by UC-03	FR-003; NFR-001
UC-12	Reschedule or Cancel Session	Extends UC-03	FR-003

6. UML Relationships

- UC-03 «include» UC-08: every booking is written against a pseudo-ID.
- UC-03 «include» UC-10: every booking checks that the selected slot is still free before commitment.
- UC-03 «include» UC-11: every confirmed booking sends a confirmation.
- UC-12 «extend» UC-03: rescheduling or cancellation occurs only when an existing booking is later changed.

7. UML Use-Case Diagram

The diagram models the actors and the use cases needed to support the selected UC-03 flow. It contains both «include» and «extend» relationships as required by the lab.

UML Use-Case Diagram — UC-03 Book Counseling Session

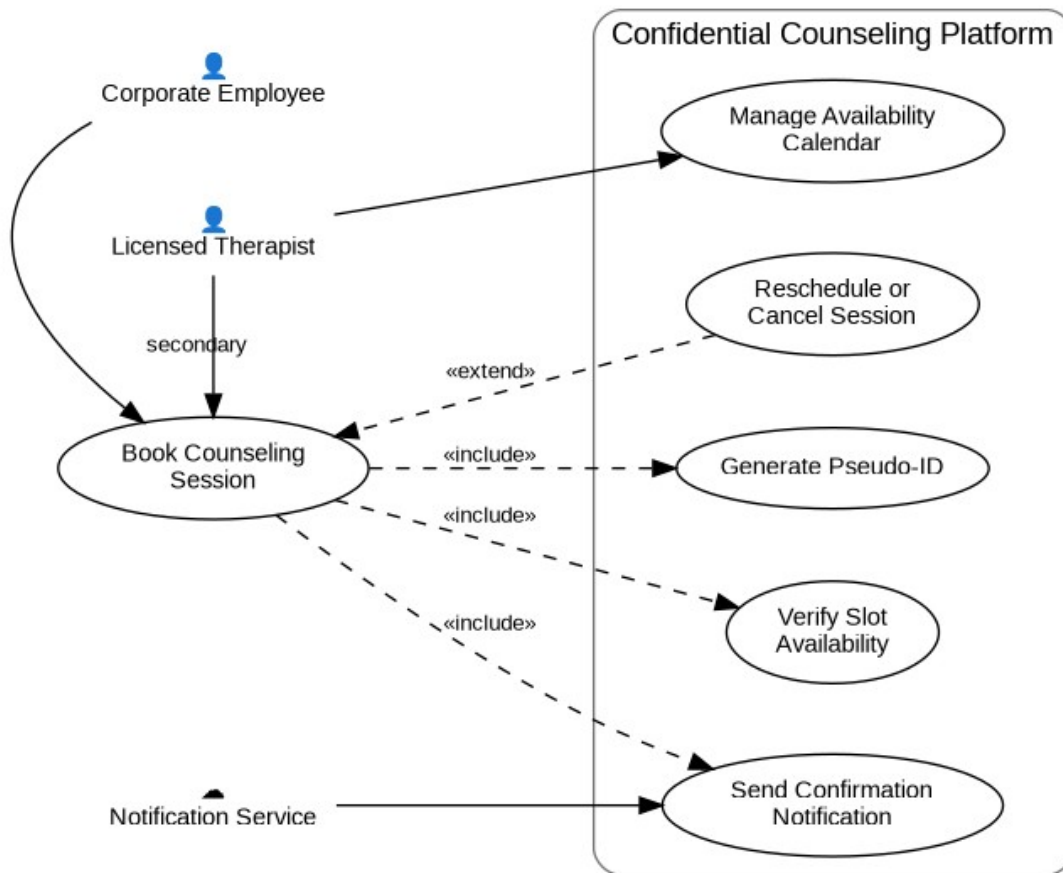


Figure 1. UML Use-Case Diagram

8. Use-Case Flow Specification — UC-03 Book Counseling Session

Primary actor: Corporate Employee

Secondary actors: Licensed Therapist (indirect), Notification Service

Goal: Reserve a confidential counselling slot with a licensed therapist without revealing the employee's corporate identity.

Trigger: The employee chooses “Book a session” from the portal.

8.1 Preconditions

1. The employee has an active account and a pseudo-ID has been issued (UC-01).
2. The employee is signed in, and the session carries only the pseudo-ID.
3. At least one licensed therapist has published availability (UC-05).

8.2 Postconditions

4. A booking record links the pseudo-ID, therapist and chosen slot.

5. The slot is marked unavailable and cannot be booked by anyone else.
6. Both parties receive a confirmation showing the time in their own timezone.
7. The therapist's view shows the pseudo-ID only.
8. Failure: no slot is reserved, therapist availability is unchanged, and the employee is told why booking did not complete.

8.3 Main Success Scenario

9. The employee selects “Book a session”.
10. The system resolves the signed-in account to its pseudo-ID («include» UC-08).
11. The employee chooses a counselling area and preferred date range.
12. The system retrieves matching published slots and displays them in the employee's local timezone.
13. The employee selects a slot and a session mode (video or chat).
14. The system re-checks that the slot is still free («include» UC-10).
15. The system reserves the slot and writes the booking against the pseudo-ID inside a single transaction.
16. The system generates the session join link and stores it against the booking.
17. The system sends confirmations to the employee and therapist («include» UC-11), each in the recipient's own timezone.
18. The system shows the employee a confirmation screen with the pseudo-ID, slot and join link.
19. The use case ends successfully.

8.4 Alternate Flow

- 6a. The selected slot was taken while the employee was choosing it.
 1. The availability check finds the slot already reserved.
 2. The system tells the employee that the slot is no longer available and does not reserve anything.
 3. The system refreshes the slot list and returns the employee to slot selection.
 4. If the employee's next two attempts also clash, the system offers the therapist's waitlist and the use case ends without a booking.

8.5 Additional Exception

- 9a. Notification service is unreachable.
 - The booking remains committed because the reservation transaction has already completed.
 - The system queues the confirmation for retry and warns the employee that email delivery is delayed.
 - The join link remains visible on the confirmation screen; delivery is retried for up to one hour before support is alerted.

9. UML Activity Diagram

Activity Diagram — Book Counseling Session

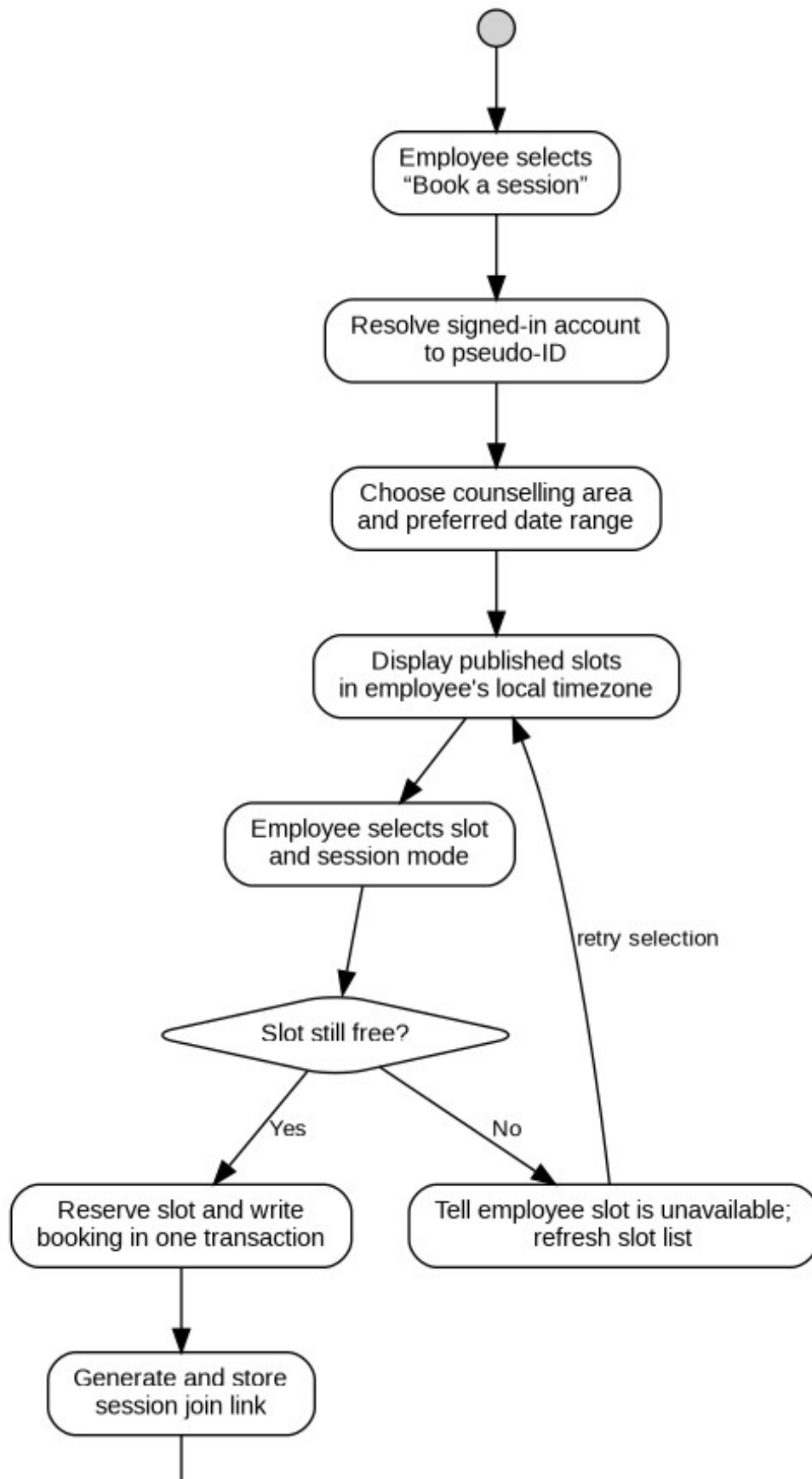


Figure 2. Activity Diagram for UC-03

10. UML Sequence Diagram

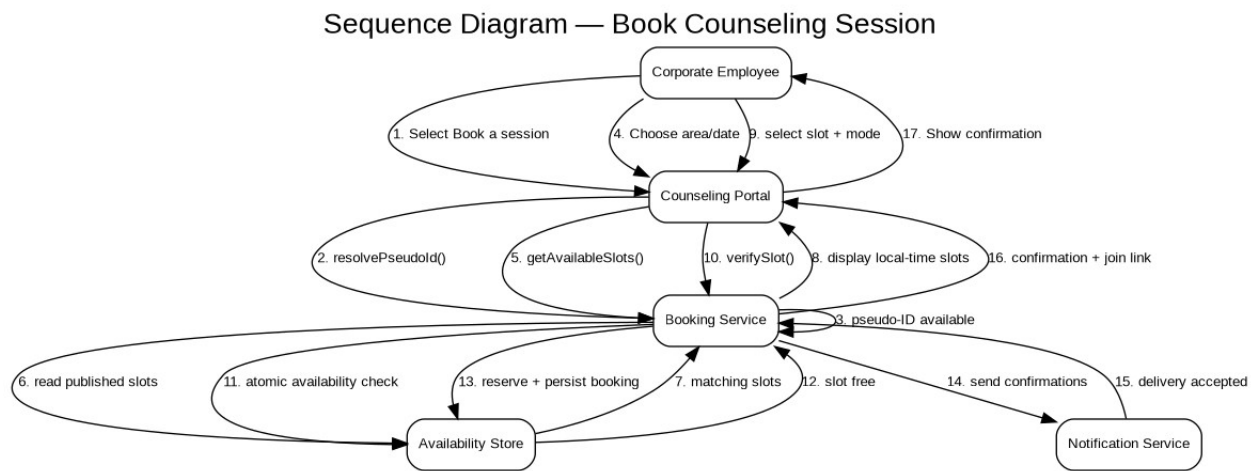


Figure 3. Sequence Diagram for UC-03

11. Scope Note

This individual report intentionally contains only one detailed use-case flow, UC-03. The other three master-report flows (UC-01, UC-02 and UC-04) have been omitted from this submission to match the instruction that each team member submit one use case report.